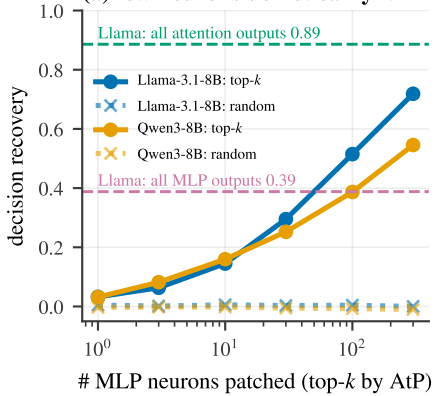
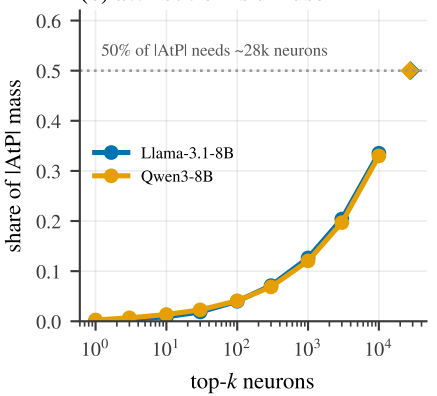
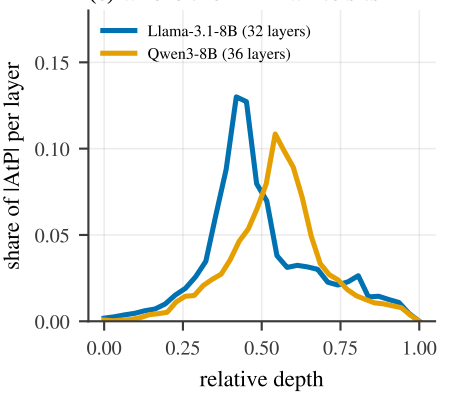
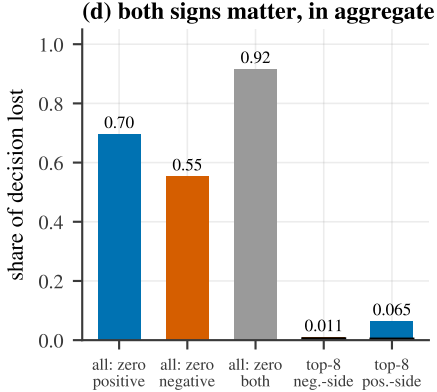
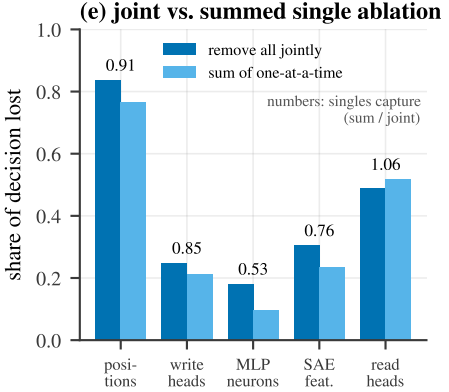
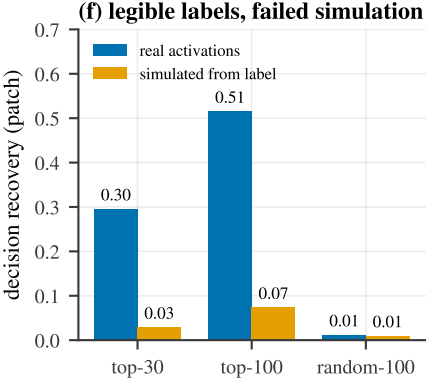

(a) few neurons do not carry it
Llama: all attention outputs 0.89
Llama-3.1-8B: top-k
Llama-3.1-8B: random
Qwen3-8B: top-k
Qwen3-8B: random
Llama: all MLP outputs 0.39
decision recovery
MLP neurons patched (top-k by AtP)


(b) attribution is diffuse
50% of |AtP| needs ~28k neurons
Llama-3.1-8B
Qwen3-8B
share of |AtP| mass
top-k neurons


(c) where the MLP write sits
Llama-3.1-8B (32 layers)
Qwen3-8B (36 layers)
share of |AtP| per layer
relative depth


(d) both signs matter, in aggregate
share of decision lost
0.70
0.55
0.92
0.011
0.065
all: zero positive
all: zero negative
all: zero both
top-8 neg.-side
top-8 pos.-side


(e) joint vs. summed single ablation
remove all jointly
sum of one-at-a-time
numbers: singles capture (sum / joint)
share of decision lost
0.91
0.85
0.53
0.76
1.06
posi-tions
write heads
MLP neurons
SAE feat.
read heads


(f) legible labels, failed simulation
real activations
simulated from label
decision recovery (patch)
0.30
0.03
0.51
0.07
0.01
0.01
top-30
top-100
random-100